2027 MOU reallocation scenarios, 15 countries (no model changes)

Generated by `audit/reallocation/run_reallocation_2027.py --year 2027` from `allocations_2027_opt_mou.csv`. Preset: America First MOUs, dashboard defaults (no government response, sudden path, death-rate trends on, 400 draws, seed 7). **US bilateral aid only:** the preset's US exits from WHO, Gavi, UNFPA and 23% of the Global Fund are switched off in memory, so the only cut is each country's MOU schedule on US direct channels (bilateral agencies, NGOs); every other donor and every multilateral channel is unchanged. Only the 2027 US MOU amount is replaced, in memory, for the 15 countries in the file; every other year and every other country keeps the schedule in `model_data/country_inputs.csv`. Totals and comparisons cover these 15 countries only.

Step 1. Data check

- `mou_deflate` is off in `model_params.csv` (amounts used nominal, as written).
- Original MOU column vs the model's `mou_us_2027`: 0 of 15 countries differ by more than 0.5% (differences of US$1 or less are rounding). Largest absolute difference: US$0.
- Column totals (US$): Original MOU 2,469,904,824.00; Best ±10% 2,500,426,243.44; Best ±20% 2,485,029,216.29; Best ±30% 2,529,398,452.76. Spread between columns: US$59,493,628.76 (NOT within US$20).

iso3	Country	Sheet Original MOU	Model mou_us_2027	Diff US$	Diff %	mou_us_ref
BWA	Botswana	37,832,582	37,832,582	0	0.000	43,742,433
BDI	Burundi	33,112,000	33,112,000	0	0.000	53,633,354
CMR	Cameroon	91,500,000	91,500,000	0	0.000	120,444,835
SWZ	Eswatini	47,000,000	47,000,000	0	0.000	62,583,783
ETH	Ethiopia	247,000,000	247,000,000	0	0.000	266,100,350
KEN	Kenya	388,294,000	388,294,000	0	0.000	406,525,908
LSO	Lesotho	55,000,000	55,000,000	0	0.000	87,734,264
LBR	Liberia	33,785,480	33,785,480	0	0.000	57,140,755
MDG	Madagascar	32,187,000	32,187,000	0	0.000	79,695,135
MWI	Malawi	165,901,540	165,901,540	0	0.000	239,128,572
MOZ	Mozambique	403,734,000	403,734,000	0	0.000	472,915,607
NGA	Nigeria	455,590,740	455,590,740	0	0.000	511,728,925
RWA	Rwanda	64,075,482	64,075,482	0	0.000	118,139,491
SLE	Sierra Leone	32,700,000	32,700,000	0	0.000	47,288,158
UGA	Uganda	382,192,000	382,192,000	0	0.000	435,160,154

Step 2. Headroom check

The preset's cut for a year is clip(1 - amount / `mou_us_ref`, 0, 1) (`scenarios.py` mou_schedule), so an amount above `mou_us_ref` is a 0% cut and the money above it buys nothing in the model (aid rises count as no loss, audit I-15). The table compares each ceiling's amount with that reference, and for information with the model's 2021-23 US baseline from `model_data/dah_lines.csv` (US direct channels, the rows the cut is applied to; and all US channels). The MOU amounts are nominal US totals from the MOU sheet and the model applies only the ratio, so only the `mou_us_ref` comparison decides whether money is lost to the clip.

iso3	Country	mou_us_ref	US direct 2021-23 (model)	US all channels 2021-23 (model)	Best ±10% 2027	Best ±20% 2027	Best ±30% 2027
BWA	Botswana	43,742,433	39,403,000	42,625,667	41,162,971	43,383,230	43,383,230
BDI	Burundi	53,633,354	18,131,000	36,964,667	24,237,693	15,363,386	6,489,079
CMR	Cameroon	120,444,835	68,536,333	109,006,000	107,316,480	119,618,186	119,618,186
SWZ	Eswatini	62,583,783	43,912,000	51,681,333	52,700,836	59,826,882	61,252,091
ETH	Ethiopia	266,100,350	104,916,333	180,462,000	264,754,170	264,754,170	264,754,170
KEN	Kenya	406,525,908	262,493,333	323,832,667	329,443,041	269,043,372	208,643,703

iso3	Country	mou_us_ref	US direct 2021-23 (model)	US all channels 2021-23 (model)	Best ±10% 2027	Best ±20% 2027	Best ±30% 2027
LSO	Lesotho	87,734,264	55,832,000	63,123,667	47,143,013	72,285,372	80,142,360
LBR	Liberia	57,140,755	18,757,333	29,447,667	24,646,534	15,507,587	55,109,688
MDG	Madagascar	79,695,135	18,203,667	37,254,333	45,320,915	32,187,000	71,588,744
MWI	Malawi	239,128,572	154,831,333	219,614,000	141,190,410	216,868,245	238,490,483
MOZ	Mozambique	472,915,607	266,907,667	365,381,000	472,835,457	472,835,457	472,835,457
NGA	Nigeria	511,728,925	269,010,667	403,669,333	510,756,355	510,756,355	510,756,355
RWA	Rwanda	118,139,491	54,217,000	87,496,000	77,149,554	90,223,625	105,476,709
SLE	Sierra Leone	47,288,158	15,085,667	28,816,000	42,103,925	45,238,566	45,238,566
UGA	Uganda	435,160,154	264,462,667	358,174,333	319,664,892	257,137,783	245,619,631

No flags: no ceiling puts any country above its `mou_us_ref` in 2027, so no money is lost to the clip; US$0 affected. (The +US$1 placeholders in the sheet raise a country's cut fraction by at most 1/ref and are treated as rounding.)

For information: 13 of the 15 Original MOU amounts already exceed the model's 2021-23 US direct-channel baseline, because `mou_us_ref` (team sheet, 2021-25 average, all US funding) is larger than the model's IHME-based US direct rows. The model never applies the dollar amounts, only cut fractions, so this is not a gain case.

Step 3. Run

- Original MOU reproduces the model's own MOU schedule for all 15 countries (cuts identical). Deaths 2026-2030 for the 15 countries are 189,629 against 356,197 in the precomputed All Countries file, which includes the US multilateral exits that are switched off here.
- 2027 cut fractions by scenario (1 - amount / mou_us_ref):

iso3	Country	Original MOU	Best ±10%	Best ±20%	Best ±30%
BWA	Botswana	13.5%	5.9%	0.8%	0.8%
BDI	Burundi	38.3%	54.8%	71.4%	87.9%
CMR	Cameroon	24.0%	10.9%	0.7%	0.7%
SWZ	Eswatini	24.9%	15.8%	4.4%	2.1%
ETH	Ethiopia	7.2%	0.5%	0.5%	0.5%
KEN	Kenya	4.5%	19.0%	33.8%	48.7%
LSO	Lesotho	37.3%	46.3%	17.6%	8.7%
LBR	Liberia	40.9%	56.9%	72.9%	3.6%
MDG	Madagascar	59.6%	43.1%	59.6%	10.2%
MWI	Malawi	30.6%	41.0%	9.3%	0.3%
MOZ	Mozambique	14.6%	0.0%	0.0%	0.0%
NGA	Nigeria	11.0%	0.2%	0.2%	0.2%
RWA	Rwanda	45.8%	34.7%	23.6%	10.7%
SLE	Sierra Leone	30.8%	11.0%	4.3%	4.3%
UGA	Uganda	12.2%	26.5%	40.9%	43.6%

Step 4. Results

4a. Totals, 15 countries

People losing a service are the model's full-cost equivalents in the year (after the 15% continuity assumption), summed over ART, TB treatment, bednets, spraying, malaria treatment and vaccines. Deaths are the central run; ranges are the 2.5th-97.5th percentiles of the draws, and the ranges of changes are paired (same draws in every scenario).

Scenario	Aid lost 2027 US$M	Aid lost 2026-30 US$M	People losing services 2027	Deaths 2026-30 (range)	Deaths in 2027	Further deaths 2031-35	HIV infections 2026-30
Original MOU	275.9	2,358.3	6,914,369	189,629 (135,102-265,790)	17,918	190,857	527,414
Best ±10%	275.9	2,358.3	5,579,943	178,058 (126,997-248,421)	13,320	187,153	505,792
Best ±20%	275.9	2,358.3	5,903,783	167,429 (119,520-230,820)	12,095	182,186	472,545
Best ±30%	275.9	2,358.3	4,795,064	162,250 (115,383-222,696)	10,837	180,274	461,450

4b. Change vs Original MOU, totals

Scenario	Aid lost 2027 US$M	Aid lost 2026-30 US$M	People losing services 2027 (range)	Deaths 2026-30 (range)	Further deaths 2031-35	HIV infections 2026-30	Deaths HIV	Deaths TB	Deaths Malaria	Deaths Immunization
Best ±10%	+0.00	+0.00	-1,334,426 (-1,460,688 to -985,948)	-11,571 (-17,409 to -7,863)	-3,705	-21,622	-7,853	-2,527	-1,007	-185
Best ±20%	-0.00	-0.00	-1,010,586 (-1,127,653 to -786,383)	-22,200 (-36,289 to -15,487)	-8,671	-54,869	-19,228	-2,187	-599	-186
Best ±30%	-0.00	-0.00	-2,119,306 (-2,318,238 to -1,535,077)	-27,379 (-44,668 to -19,312)	-10,583	-65,964	-23,552	-1,872	-1,746	-210

- **Best ±10%:** the reallocation is not zero-sum in outcomes: 11,571 fewer deaths over 2026-2030 (-6.10% of the Original MOU total), 1,334,426 fewer people losing a service in 2027, and +0.00 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).
- **Best ±20%:** the reallocation is not zero-sum in outcomes: 22,200 fewer deaths over 2026-2030 (-11.71% of the Original MOU total), 1,010,586 fewer people losing a service in 2027, and -0.00 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).
- **Best ±30%:** the reallocation is not zero-sum in outcomes: 27,379 fewer deaths over 2026-2030 (-14.44% of the Original MOU total), 2,119,306 fewer people losing a service in 2027, and -0.00 US$M of modelled aid lost in 2027 (the same nominal total of MOU money buys a different amount of modelled aid because each country's cut fraction is applied to its own 2021-23 US direct-channel baseline).

4c. People losing each service in 2027 and coverage change (totals)

Service	Original MOU people	Original MOU coverage drop pp	Best ±10% Δ people	Best ±10% Δ pp	Best ±20% Δ people	Best ±20% Δ pp	Best ±30% Δ people	Best ±30% Δ pp
ART	455,448	4.09	-30,031	-0.270	-95,133	-0.855	-83,363	-0.749
TB treatment	7,752	0.62	-5,222	-0.416	-4,194	-0.334	-3,240	-0.258
Bednets	5,595,288	0.94	-1,206,940	-0.203	-914,134	-0.154	-1,877,225	-0.317
Spraying	464,390	0.08	-49,118	-0.008	+5,594	+0.001	-76,922	-0.013
Malaria treatment	382,000	0.27	-37,663	-0.027	+2,414	+0.002	-73,083	-0.052
Vaccines	9,491	0.05	-5,452	-0.026	-5,133	-0.024	-5,472	-0.026
PMTCT	9,440	2.45	+2,456	+0.637	+5,154	+1.338	+7,063	+1.833
HIV prevention (infections averted)	4,529	n/a	-279	n/a	-1,095	n/a	-1,204	n/a

4d. Deaths by disease 2026-2030, totals (range), and the part caused by the 2027 change

Disease	Original MOU deaths 2026-30 (range)	Best ±10% Δ (paired range)	Best ±20% Δ (paired range)	Best ±30% Δ (paired range)
HIV	118,793 (78,587-201,179)	-7,853 (-13,529 to -4,893)	-19,228 (-33,236 to -12,802)	-23,552 (-40,492 to -15,981)
TB	33,803 (20,050-45,231)	-2,527 (-3,417 to -1,457)	-2,187 (-2,942 to -1,318)	-1,872 (-2,509 to -1,145)
Malaria	34,738 (18,948-39,878)	-1,007 (-1,167 to -615)	-599 (-855 to -336)	-1,746 (-2,036 to -1,066)
Immunization	2,296 (1,689-4,633)	-185 (-354 to -136)	-186 (-352 to -138)	-210 (-398 to -155)

Disease	Original MOU deaths 2026-30 (range)	Best ±10% Δ (paired range)	Best ±20% Δ (paired range)	Best ±30% Δ (paired range)
All	189,629 (135,102-265,790)	-11,571 (-17,409 to -7,863)	-22,200 (-36,289 to -15,487)	-27,379 (-44,668 to -19,312)

Deaths by calendar year, change vs Original MOU (central): the 2027 change also moves deaths in later years (ART cohorts age, TB cases spread 1-2 years later, missed vaccine cohorts die over following years).

Scenario	2026	2027	2028	2029	2030	2031-35
Best ±10%	+0	-4,597	-2,354	-2,377	-2,243	-3,705
Best ±20%	+0	-5,823	-5,415	-5,527	-5,435	-8,671
Best ±30%	+0	-7,080	-6,699	-6,838	-6,761	-10,583

4e. By country: 2027 money change, change in deaths 2026-30 and in people losing services in 2027

Nominal US$ moved is the sheet's change in the country's 2027 MOU amount; modelled aid is the change in the model's aid lost in 2027 (cut fraction x the country's US direct-channel baseline). The per-US$1M columns divide the change in deaths by the money moved, so they are positive in both directions: lives saved per US$1M where money arrives, lives lost per US$1M where it leaves.

Best ±10% minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+3.33	+3.00	-461 (-839 to -283)	-214	-4,572	-3,663	+138.3	+153.6
BDI	Burundi	-8.87	-3.00	+125 (+92 to +211)	+41	+53,174	+326	+14.0	+41.5
CMR	Cameroon	+15.82	+9.00	-1,584 (-2,670 to -1,009)	-681	-357,567	-3,609	+100.1	+176.0
SWZ	Eswatini	+5.70	+4.00	-843 (-1,424 to -443)	-309	-8,182	-2,318	+147.9	+210.8
ETH	Ethiopia	+17.75	+7.00	-1,083 (-1,674 to -806)	-211	-190,847	-883	+61.0	+154.7
KEN	Kenya	-58.85	-38.00	+2,987 (+1,912 to +4,885)	+853	+624,690	+4,697	+50.8	+78.6
LSO	Lesotho	-7.86	-5.00	+295 (+214 to +533)	+190	+10,309	+2,716	+37.5	+59.0
LBR	Liberia	-9.14	-3.00	+134 (+86 to +152)	+17	+133,014	+103	+14.7	+44.8
MDG	Madagascar	+13.13	+3.00	-293 (-310 to -142)	-1	-288,759	-3	+22.3	+97.7
MWI	Malawi	-24.71	-16.00	+1,148 (+832 to +1,820)	+392	+263,345	+3,255	+46.4	+71.7
MOZ	Mozambique	+69.10	+39.00	-7,310 (-11,958 to -5,011)	-2,381	-398,841	-16,721	+105.8	+187.4
NGA	Nigeria	+55.17	+29.00	-5,698 (-8,032 to -3,790)	-1,751	-1,342,539	-10,078	+103.3	+196.5
RWA	Rwanda	+13.07	+6.00	-869 (-1,311 to -652)	-290	-198,749	-1,965	+66.5	+144.8
SLE	Sierra Leone	+9.40	+3.00	-451 (-596 to -301)	-91	-249,423	-597	+48.0	+150.4

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
UGA	Uganda	-62.53	-38.00	+2,332 (+1,503 to +3,537)	+731	+620,520	+7,117	+37.3	+61.4
TOTAL	All 15	0.00	-0.00	-11,571 (-17,409 to -7,863)	-3,705	-1,334,426	-21,622		

Best ±20% minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+5.55	+5.00	-768 (-1,399 to -471)	-357	-7,620	-6,104	+138.3	+153.6
BDI	Burundi	-17.75	-6.00	+193 (+142 to +312)	+54	+106,347	+549	+10.8	+32.1
CMR	Cameroon	+28.12	+16.00	-3,260 (-5,537 to -2,067)	-1,351	-635,675	-7,104	+115.9	+203.7
SWZ	Eswatini	+12.83	+9.00	-1,897 (-3,204 to -998)	-696	-18,411	-5,216	+147.9	+210.8
ETH	Ethiopia	+17.75	+7.00	-1,083 (-1,674 to -806)	-211	-190,847	-883	+61.0	+154.7
KEN	Kenya	-119.25	-77.00	+4,453 (+2,894 to +6,978)	+1,006	+1,265,819	+7,777	+37.3	+57.8
LSO	Lesotho	+17.29	+11.00	-2,201 (-4,042 to -1,536)	-1,178	-22,680	-12,749	+127.3	+200.1
LBR	Liberia	-18.28	-6.00	+257 (+162 to +289)	+27	+266,029	+182	+14.1	+42.8
MDG	Madagascar	+0.00	-0.00	+0 (+0 to +0)	+0	+0	+0	n/a	n/a
MWI	Malawi	+50.97	+33.00	-6,398 (-10,701 to -4,542)	-2,175	-543,149	-12,313	+125.5	+193.9
MOZ	Mozambique	+69.10	+39.00	-7,310 (-11,958 to -5,011)	-2,381	-398,841	-16,721	+105.8	+187.4
NGA	Nigeria	+55.17	+29.00	-5,698 (-8,032 to -3,790)	-1,751	-1,342,539	-10,078	+103.3	+196.5
RWA	Rwanda	+26.15	+12.00	-1,737 (-2,620 to -1,303)	-579	-397,498	-3,931	+66.4	+144.7
SLE	Sierra Leone	+12.54	+4.00	-614 (-817 to -409)	-125	-332,563	-819	+49.0	+153.5
UGA	Uganda	-125.05	-76.00	+3,863 (+2,525 to +5,623)	+1,046	+1,241,040	+12,541	+30.9	+50.8
TOTAL	All 15	0.00	+0.00	-22,200 (-36,289 to -15,487)	-8,671	-1,010,586	-54,869		

Best ±30% minus Original MOU

iso3	Country	MOU US$M moved	Modelled aid US$M (+ = more aid)	Δ deaths 2026-30 (range)	Δ deaths 2031-35	Δ people losing services 2027	Δ HIV infections	Deaths per US$1M moved (nominal; + = lives the money saves where it arrives, or costs where it leaves)	per US$1M modelled
BWA	Botswana	+5.55	+5.00	-768 (-1,399 to -471)	-357	-7,620	-6,104	+138.3	+153.6
BDI	Burundi	-26.62	-9.00	+260 (+194 to +414)	+66	+159,521	+772	+9.8	+28.9
CMR	Cameroon	+28.12	+16.00	-3,260 (-5,537 to -2,067)	-1,351	-635,675	-7,104	+115.9	+203.7
SWZ	Eswatini	+14.25	+10.00	-2,107 (-3,560 to -1,109)	-773	-20,456	-5,796	+147.9	+210.7
ETH	Ethiopia	+17.75	+7.00	-1,083 (-1,674 to -806)	-211	-190,847	-883	+61.0	+154.7
KEN	Kenya	-179.65	-116.00	+5,921 (+3,868 to +8,956)	+1,159	+1,906,948	+10,856	+33.0	+51.0
LSO	Lesotho	+25.14	+16.00	-3,289 (-6,043 to -2,293)	-1,748	-32,990	-18,897	+130.8	+205.6
LBR	Liberia	+21.32	+7.00	-464 (-601 to -321)	-106	-310,367	-485	+21.7	+66.3
MDG	Madagascar	+39.40	+9.00	-873 (-925 to -425)	-2	-866,278	-8	+22.2	+97.0
MWI	Malawi	+72.59	+47.00	-9,491 (-15,896 to -6,739)	-3,191	-773,576	-18,014	+130.8	+201.9
MOZ	Mozambique	+69.10	+39.00	-7,310 (-11,958 to -5,011)	-2,381	-398,841	-16,721	+105.8	+187.4
NGA	Nigeria	+55.17	+29.00	-5,698 (-8,032 to -3,790)	-1,751	-1,342,539	-10,078	+103.3	+196.5
RWA	Rwanda	+41.40	+19.00	-2,749 (-4,147 to -2,062)	-917	-629,371	-6,224	+66.4	+144.7
SLE	Sierra Leone	+12.54	+4.00	-614 (-817 to -409)	-125	-332,563	-819	+49.0	+153.5
UGA	Uganda	-136.57	-83.00	+4,145 (+2,709 to +6,007)	+1,104	+1,355,346	+13,540	+30.4	+49.9
TOTAL	All 15	0.00	+0.00	-27,379 (-44,668 to -19,312)	-10,583	-2,119,306	-65,964		

Step 5. Efficiency: the marginal effect of US$1M of 2027 US MOU funding

Method: for each country, the Original MOU run minus a run with the country's 2027 MOU amount lowered by US$1M (central parameters, everything else unchanged). 'People kept' are the extra full-cost equivalents on each service in 2027; 'deaths averted' are the extra deaths over 2026-2030 (and 2031-2035) that the US$1M prevents. A +US$1M run checks linearity. The model's own US$1M cards (`health_model.per_million_card`, audit I-04: a uniform US$1M-a-year cut for five years to one disease across all donors and channels) are shown afterwards for comparison; they are per disease, per year of money, and not scaled by the MOU leverage, so they are not the same quantity.

5a. Ranking by deaths averted per US$1M of 2027 MOU money (nominal)

Rank	Country	Leverage (model $ per MOU $)	Modelled aid per US$1M, US$M	ART kept	TB kept	Bednets kept	Spraying kept	Malaria Rx kept	Vaccines kept	All six services kept	Deaths averted 2026-30 per US$1M	Further 2031-35	Deaths averted per US$1M modelled aid	Check: +US$1M side	HIV infections averted
1	Botswana	0.90	0.90	1,366	0	6	1	0	0	1,373	138.3	64.3	153.6	138.3	1099.7
2	Nigeria	0.53	0.53	1,075	46	21,388	794	984	49	24,337	93.9	28.9	178.5	93.9	167.4

Rank	Country	Leverage (model $ per MOU $)	Modelled aid per US$1M, US$M	ART kept	TB kept	Bednets kept	Spraying kept	Malaria Rx kept	Vaccines kept	All six services kept	Deaths averted 2026-30 per US$1M	Further 2031-35	Deaths averted per US$1M modelled aid	Check: +US$1M side	HIV infections averted
3	Mozambique	0.56	0.56	1,059	47	3,730	611	297	29	5,772	82.6	27.8	146.3	82.6	201.2
4	Eswatini	0.70	0.70	1,419	8	7	2	0	0	1,435	79.3	35.5	113.0	147.9	277.8
5	Malawi	0.65	0.65	1,379	2	7,845	618	807	5	10,657	77.0	30.7	119.0	77.0	180.5
6	Cameroon	0.57	0.57	1,254	5	20,361	304	669	14	22,607	75.5	35.3	132.6	75.5	190.0
7	Lesotho	0.64	0.64	1,299	1	9	2	1	0	1,312	73.7	47.2	115.9	73.7	523.9
8	Rwanda	0.46	0.46	597	0	12,443	1,741	416	5	15,202	66.5	22.1	144.9	66.5	150.3
9	Ethiopia	0.39	0.39	381	44	7,743	1,366	1,184	33	10,749	61.0	11.9	154.7	61.0	49.7
10	Kenya	0.65	0.65	914	16	8,784	314	572	15	10,615	57.4	17.5	88.8	57.4	87.0
11	Uganda	0.61	0.61	801	6	6,941	1,346	827	3	9,924	53.6	20.1	88.3	53.6	148.3
12	Sierra Leone	0.32	0.32	272	0	23,056	1,809	1,257	130	26,523	39.3	7.1	123.1	39.2	47.5
13	Madagascar	0.23	0.23	0	0	18,379	1,630	1,972	4	21,986	22.4	0.0	98.0	22.4	0.2
14	Burundi	0.34	0.34	324	0	4,873	573	220	1	5,992	19.5	7.3	57.6	19.5	46.6
15	Liberia	0.33	0.33	111	0	13,462	452	506	24	14,555	17.3	3.4	52.6	17.3	16.1

5b. Drivers

Leverage = the country's 2021-23 US direct-channel aid in the model / `mou_us_ref` : how many modelled dollars one nominal MOU dollar moves. Disease mix = share of the country's US direct-channel aid by disease (the cut is uniform across the US direct rows, so the money lost follows this mix). Service money per US$1M = the year's extra loss landing on each service line after the model splits unspecified and systems money (`health_model.run_country` effective()). Unit costs are the central values the model uses (2023 US$). Deaths per unit = extra deaths 2026-2030 per extra person losing the service in 2027.

Disease mix of US direct aid and service money per US$1M of MOU money (US$ thousands by line; the rest is non-service money: programme management, systems, OVC and testing money beyond the kappa share)

Country	Leverage	US mix HIV	US mix TB	US mix Malaria	US mix Immunization	ART	HIV testing & care	HIV prevention	PMTCT	OVC	TB DS	Bednets	IRS	Malaria Rx	Vaccines
Botswana	0.90	99%	0%	0%	0%	254	161	229	3	17	2	0	0	0	0
Nigeria	0.53	73%	7%	19%	1%	203	120	20	5	15	13	55	8	16	5
Mozambique	0.56	82%	11%	7%	1%	229	109	70	10	7	27	13	6	5	3
Eswatini	0.70	95%	5%	0%	0%	261	116	99	6	36	21	0	0	0	0
Malawi	0.65	89%	2%	9%	0%	297	51	94	5	9	4	20	6	13	2
Cameroon	0.57	78%	2%	20%	0%	278	82	22	3	19	3	53	3	11	1
Lesotho	0.64	99%	0%	0%	0%	199	185	95	12	28	1	0	0	0	0
Rwanda	0.46	78%	0%	21%	0%	170	34	58	5	14	0	32	17	15	1
Ethiopia	0.39	67%	10%	23%	1%	103	76	17	1	15	20	35	14	20	3
Kenya	0.65	87%	4%	8%	0%	249	165	62	12	16	5	23	3	9	2
Uganda	0.61	84%	6%	10%	0%	243	106	75	14	17	3	18	13	14	0
Sierra Leone	0.32	37%	1%	57%	4%	48	27	18	0	0	0	60	18	21	13
Madagascar	0.23	3%	2%	95%	0%	0	0	1	0	0	0	48	16	32	0
Burundi	0.34	68%	0%	32%	0%	78	72	9	7	11	0	13	6	4	0
Liberia	0.33	39%	4%	57%	1%	29	47	5	1	0	0	35	5	8	2

Unit costs (US$ per person-year, course, net, person sprayed, case treated, child) and health effect per person losing the service

Country	ART $/py	TB DS $/course	Bednet $	IRS $/person	Malaria Rx $/case	Vaccines $/child	Deaths 2026-30 per ART py lost	per TB course lost	per 1,000 bednets lost	per 1,000 malaria cases untreated	per 1,000 children unvaccinated	HIV transmission /unsuppressed py	ART coverage	TB-HIV share	Malaria deaths per 1,000 cases	Bednet use	U5MR	Vaccine deaths averted per 1,000 children
Botswana	208	1,232	2.20	8.50	14.00	87	0.1001	0.491	0.000	n/a	n/a	0.308	100%	44%	0.92	50%	33	33.3
Nigeria	208	246	2.20	8.50	14.00	87	0.0510	0.454	0.495	4.76	40.75	0.122	78%	5%	2.68	28%	116	59.0
Mozambique	227	500	2.20	8.50	14.00	87	0.0507	0.468	0.327	2.97	22.10	0.130	84%	23%	1.79	43%	59	30.4
Eswatini	191	246	2.20	8.50	14.00	87	0.0511	0.527	0.000	0.00	n/a	0.137	95%	55%	0.00	50%	45	34.1
Malawi	199	500	2.20	8.50	14.00	87	0.0504	0.498	0.267	2.36	20.37	0.098	89%	46%	1.21	59%	49	27.8
Cameroon	216	246	2.20	8.50	14.00	87	0.0512	0.485	0.281	1.93	26.37	0.118	73%	16%	1.29	48%	65	39.4
Lesotho	191	246	2.20	8.50	14.00	87	0.0535	0.535	0.000	0.00	26.73	0.251	96%	58%	n/a	50%	60	36.3
Rwanda	208	500	2.20	8.50	14.00	87	0.0995	0.427	0.133	9.97	13.05	0.126	94%	12%	6.10	42%	38	18.9
Ethiopia	314	397	2.20	8.50	14.00	87	0.0975	0.453	0.138	1.57	23.25	0.073	82%	6%	1.34	9%	44	34.9
Kenya	308	273	2.20	8.50	14.00	87	0.0498	0.337	0.064	5.64	13.99	0.070	91%	21%	3.03	44%	39	20.9
Uganda	314	500	2.20	8.50	14.00	87	0.0506	0.462	0.251	3.51	18.73	0.125	85%	33%	1.35	68%	49	26.9
Sierra Leone	191	500	2.20	8.50	14.00	87	0.0508	0.475	0.637	5.29	28.79	0.132	84%	14%	3.04	83%	90	51.3
Madagascar	400	500	2.20	8.50	14.00	87	0.0527	0.469	0.472	6.65	16.45	0.120	24%	4%	5.15	42%	62	24.8
Burundi	299	500	2.20	8.50	14.00	87	0.0505	0.467	0.296	2.40	22.82	0.110	89%	6%	1.31	40%	47	29.6
Liberia	400	500	2.20	8.50	14.00	87	0.0514	0.480	0.498	7.30	34.08	0.111	59%	10%	3.48	61%	86	64.8

Caps and nonlinearity:

- Lines capped at the people covered (Original MOU run): none.
- Lines where donor money buys more than the donor-funded pool, so units scale with the share of money lost (`donor_exceeds_cost`): Botswana tb_ds; Botswana tb_dr; Botswana mal_cm; Botswana imm; Nigeria hiv_pmtct; Mozambique tb_dr; Mozambique mal_itn; Eswatini tb_ds; Eswatini tb_dr; Eswatini mal_itn; Eswatini mal_cm; Eswatini imm; Malawi tb_ds; Malawi tb_dr; Malawi imm; Cameroon hiv_pmtct; Cameroon tb_ds; Cameroon tb_dr; Lesotho hiv_pmtct; Lesotho tb_ds; Lesotho tb_dr; Rwanda hiv_art; Rwanda hiv_pmtct; Rwanda tb_ds; Rwanda tb_dr; Rwanda mal_cm; Ethiopia hiv_pmtct; Ethiopia tb_dr; Ethiopia mal_itn; Kenya hiv_pmtct; Uganda tb_dr; Uganda imm; Madagascar hiv_pmtct; Burundi hiv_pmtct; Burundi tb_ds; Burundi tb_dr; Liberia hiv_pmtct.
- Countries where the -US$1M and +US$1M marginals differ by more than 2% (nonlinear: malaria Lives Saved Tool, ART cohort ageing, TB spread): Eswatini (79.27 vs 147.88).

5c. The model's own US$1M cards (audit I-04), for comparison

Deaths 2026-2030 from a US$1M-a-year loss for five years to one disease (all donors and channels), and the main service's people losing it per year (HIV: ART; TB: TB treatment; malaria: bednets; immunization: routine vaccines).

Country	HIV deaths per US$1M/yr	TB deaths per US$1M/yr	Malaria deaths per US$1M/yr	Immunization deaths per US$1M/yr	HIV people per US$1M/yr	TB people per US$1M/yr	Malaria people per US$1M/yr	Immunization people per US$1M/yr
Botswana	273.3	219.5	2.1	0.0	1,392	96	79,592	0
Nigeria	425.7	2703.7	765.0	1873.6	2,144	1,303	161,507	9,770
Mozambique	384.0	1741.9	393.1	1016.1	1,788	805	107,413	9,770
Eswatini	364.5	409.3	0.0	436.9	1,827	168	39,625	3,997
Malawi	414.9	693.8	411.5	307.0	2,083	303	136,367	3,181
Cameroon	437.8	1529.2	342.1	1204.8	2,067	689	143,618	9,770
Lesotho	385.6	1004.3	0.0	1233.9	1,899	410	65,519	9,770
Rwanda	261.0	465.7	435.4	595.4	1,304	231	129,543	9,770

Country	HIV deaths per US$1M/yr	TB deaths per US$1M/yr	Malaria deaths per US$1M/yr	Immunization deaths per US$1M/yr	HIV people per US$1M/yr	TB people per US$1M/yr	Malaria people per US$1M/yr	Immunization people per US$1M/yr
Ethiopia	222.4	2335.0	203.5	1063.4	1,099	1,118	78,353	9,770
Kenya	290.2	2559.9	580.8	644.2	1,364	1,204	150,197	9,770
Uganda	292.9	1274.3	497.1	469.5	1,285	586	125,121	5,320
Sierra Leone	336.3	1559.0	687.3	1312.0	1,543	707	138,408	9,770
Madagascar	131.5	1440.6	709.2	749.5	573	661	101,837	9,770
Burundi	186.1	1660.6	309.5	1049.2	912	765	96,365	9,770
Liberia	159.8	1464.0	554.5	1555.4	723	647	100,811	9,770

Step 5 (continued). Explanation

How to read this report

The three scenarios are the best allocations of the 2027 US MOU money for deaths found by `audit/reallocation/run_optimal_realloc_2027.py` (RESULTS_optimal_2027.md), with each country's change limited to ±10%, ±20% or ±30% of its signed 2027 amount. They are defined **dollar for dollar**: US$1 taken from one country is US$1 of US direct aid given to another, so the model's aid lost in 2027 is unchanged (tables 4a-4b: +0.00 US$M), and no country is raised above its 2021-23 US aid level (a 0% cut; the model has no coverage above baseline) or cut below zero US aid.

This report's script works in MOU dollars and turns them into cut fractions, so each allocation was converted into the MOU amount that produces exactly that aid change in the model (`allocations_2027_opt_mou.csv`; MOU amount = ref x (1 - (Original cut - aid change / US direct baseline)); the aid change matches to within US$0.01). Because one MOU dollar moves a different amount of modelled aid in each country (Step 2 and table 5a), these MOU-equivalent amounts do not add up to the Original total: Step 1 flags the spread (US$2,500M / 2,485M / 2,529M against US$2,469.9M) and that is expected, not an error. In table 4e, the "MOU US$M moved" column is this MOU-equivalent amount; **the "Modelled aid US$M" column is the real dollar change**, and it sums to zero. (The TOTAL row prints 0.00 for MOU dollars moved by construction; the MOU-equivalent amounts sum to +US$30.5M / +US$15.1M / +US$59.5M.)

The actual dollar changes (US$M of US aid, face value):

Country	±10%	±20%	±30%	Original 2027 cut
Botswana	+3.0	+5.0	+5.0	13.5%
Burundi	-3.0	-6.0	-9.0	38.3%
Cameroon	+9.0	+16.0	+16.0	24.0%
Eswatini	+4.0	+9.0	+10.0	24.9%
Ethiopia	+7.0	+7.0	+7.0	7.2%
Kenya	-38.0	-77.0	-116.0	4.5%
Lesotho	-5.0	+11.0	+16.0	37.3%
Liberia	-3.0	-6.0	+7.0	40.9%
Madagascar	+3.0	0.0	+9.0	59.6%
Malawi	-16.0	+33.0	+47.0	30.6%
Mozambique	+39.0	+39.0	+39.0	14.6%
Nigeria	+29.0	+29.0	+29.0	11.0%
Rwanda	+6.0	+12.0	+19.0	45.8%
Sierra Leone	+3.0	+4.0	+4.0	30.8%
Uganda	-38.0	-76.0	-83.0	12.2%
Moved	**103**	**165**	**208**	

What drives the result (numbers from tables 4b-4e and 5a)

1. **Money goes where restoring a cut saves the most lives per dollar.** The receivers are mostly HIV-heavy countries whose Original MOU cuts US aid deeply (Rwanda 46%, Lesotho 37%, Malawi 31%, Eswatini 25%, Cameroon 24%) or whose aid buys treatment cheaply (Nigeria, Mozambique: ART US$208-227 per patient-year). For these HIV-heavy receivers, restoring that money averts 145-211 deaths per US$1M in the ±30% scenario (table 4e, per US$1M modelled: Eswatini 211, Lesotho 206, Cameroon 204, Malawi 202, Nigeria 197, Mozambique 187), more than the marginal US$1M at the Original level suggests (Malawi 119, table 5a): when a 2027 cut is reversed, people who stopped ART in 2026 return to care, so each dollar restored first brings back people already at high risk.
2. **Money comes from where a cut costs the fewest lives per dollar.** Kenya and Uganda: ART costs US$308-314 per patient-year, a fifth to almost a third of their HIV money goes to testing and care with no modelled deaths (Uganda 21%, Kenya 29%), and their Original cuts are shallow (4.5% and 12.2%), so cutting further costs about 50 deaths per US$1M (table 4e: Kenya 51.0, Uganda 49.9). Burundi (29 per US$1M) gives in every scenario.
3. **The curves are not straight lines, so the allocation changes shape with the bound.** Lesotho and Malawi give money at ±10% but receive at ±20% and ±30%; Liberia gives at ±10-20% and receives at ±30%. Their first dollars restored are worth less than later ones (the 2026 cohort returns only once the 2027 cut falls below the 2026 cut), so they only pay off when enough money can be moved.
4. **Everything improves at once.** Unlike the team's ceilings, the best allocations reduce deaths in every disease (±30%: HIV -23,552, TB -1,872, malaria -1,746, immunization -210), avert 21,622 / 54,869 / 65,964 HIV infections and keep 1.0-2.1 million more people on a service in 2027 (table 4b). Deaths fall most in 2027 itself (-4,597 / -5,823 / -7,080) and the gain persists to 2035 (table 4d, by calendar year).
5. **Ranges.** The paired ranges of the change in deaths are well below zero in all three scenarios (±10% -17,409 to -7,863; ±20% -36,289 to -15,487; ±30% -44,668 to -19,312), so the direction does not depend on the uncertain parameters.

Who pays for it

The gains are paid for by Kenya, Uganda and Burundi. In the ±30% scenario ART coverage in 2027 falls from 90.0% to 78.9% in Kenya (164,000 more people off treatment), from 82.4% to 75.4% in Uganda (109,000) and from 81.3% to 70.8% in Burundi, while Malawi returns from 79.0% to 89.0%, Lesotho from 79.5% to 92.3%, Eswatini from 84.9% to 94.3% and Rwanda from 80.5% to 91.0% (RESULTS_optimal_2027.md, section 3). Kenya and Uganda alone add 10,066 deaths over 2026-2030 (table 4e); the other twelve countries avert 37,445.

In plain language

1. Moving 4-8% of the 2027 US MOU money between the same 15 countries, with no new money, could avert 11,600 to 27,400 deaths over 2026-2030 in the model (6-14% of the 189,629 under the Original MOU), and every disease improves.
2. The money should go back to countries whose US HIV aid is cut deeply or buys treatment cheaply (Malawi, Mozambique, Nigeria, Rwanda, Lesotho, Cameroon, Eswatini), and come mainly from Kenya and Uganda, where treatment is expensive and the Original cuts are small.
3. This is the model's answer to "fewest deaths in total". It is not neutral between countries: Kenya, Uganda and Burundi lose 7-11 points of ART coverage in the widest version. A fairness limit (for example, no country's ART coverage more than a few points below its Original MOU level) would keep part of the gain.
4. Limits: these are the model's numbers (unit costs, the ART cohort rule, central parameters for the allocation itself); the allocation is the best one found on US$1M steps, not a proven optimum; and each MOU is a separate bilateral agreement, so this shows where money does most in the model, not what is negotiable.